Documentation: Customer Shopping Behavior Analysis and Purchase Amount Prediction

1. Overview of the Project

This project aimed to segment the customer base of an e-commerce platform based on their shopping behavior and subsequently build a classification model to predict a customer's Purchase Segment (Low, Medium, or High) using their demographic and transactional features.

The analysis was executed in two main stages:

1. **Customer Segmentation (Unsupervised Learning):** Using the K-Means algorithm on preprocessed customer data.
2. **Purchase Segment Classification (Supervised Learning):** Training Logistic Regression and Random Forest models to predict the purchase segment, using the identified customer cluster as an additional feature.

2. Data Preparation and Preprocessing

The initial dataset (shopping_behavior_updated.csv) contained 3900 entries with 18 columns, including a mix of numerical and categorical features (e.g., Age, Purchase Amount (USD), Gender, Category). This is one of the most crucial stages, as the performance of any Machine Learning model depends directly on the quality and format of the input data. In this case, the objective was to prepare the data for the **K-Means** segmentation algorithm.

- **Missing Values:** No missing values were found in the dataset. The dataset was **complete** (no null or missing values that needed imputation or removal).

- *2.2. The Need for Preprocessing (Scaling and Encoding)*

The **K-Means** algorithm works by calculating distances between data points. For this distance calculation to be accurate and fair, two conditions must be met:

- **All features must be numeric.** (K-Means cannot process text/categories).
    - **Preprocessing Pipeline:** A ColumnTransformer pipeline was created for mandatory feature scaling and encoding:
        - **Numerical Features** (Age, Purchase Amount (USD), Previous Purchases, Review Rating): Scaled using StandardScaler.
        - **Categorical Features:** Encoded using OneHotEncoder (binary values to each quality) to convert them into a numerical format suitable for the K-Means algorithm.
- **All numeric features must be on a similar scale.** (If Purchase Amount ranges from 1 to 1000 and Age ranges from 18 to 60, the purchase amount would unfairly dominate the distance calculation).

- *2.2.1 The Role of ColumnTransformer:*

- **Numerical Features** (e.g., Age, Purchase Amount): These need **scaling** so that high-magnitude values don't dominate the distance calculations.
- **Categorical Features** (e.g., Gender, Location): These are text-based and must be converted into **numerical format** using encoding.

Without ColumnTransformer, you would have to manually separate the columns, transform each group separately, and then recombine them—a process that is error-prone and hard to maintain.The ColumnTransformer takes a list of tuples, where each tuple specifies three components:

Component	Description	Example from Your Code
Name	A unique string identifier for the transformation step.	`'num'` or `'cat'`
Transformer	The specific preprocessing object (e.g., `StandardScaler` or `OneHotEncoder`).	`StandardScaler()` or `OneHotEncoder()`
Columns	A list of column names (features) to which the transformer should be applied.	`numerical_features` or `categorical_features`

- StandarScaler(): This transformation scales the data so that it has a mean of 0 and a standard deviation of 1. This prevents features with large magnitudes (like Purchase Amount) from dominating the distance calculations in K-Means.
- OneHotEncoder(): categorical into binary column.

- *2.2.2 Combining Transformations with ColumnTransformer*

The Brain of Preprocessing: The ColumnTransformer is configured to apply the defined pipelines to the correct feature subsets:

- The pipeline named 'num' (containing StandardScaler) is applied to numerical_features.
- The pipeline named 'cat' (containing OneHotEncoder) is applied to categorical_features.

remainder='passthrough': This specifies that any columns not explicitly mentioned in the lists (if any were left over) should be kept unchanged in the output.

3. Customer Segmentation (K-Means)

- 3.1 K-Means

K-Means Segmentation (or Clustering) is a foundational unsupervised machine learning algorithm. Its primary purpose is not to predict an outcome (as a supervised model would), but rather to discover and group hidden patterns within a dataset.

The core idea of K-Means is to partition a dataset into K distinct groups (where K is a number you pre-define) such that:

1. **High Intra-Cluster Similarity (Cohesion):** Data points within the same cluster are as similar as possible to each other.
2. **Low Inter-Cluster Similarity (Separation):** Data points from different clusters are as dissimilar as possible.

The preprocessed data (X_processed) was used to find natural groupings in the customer base.

3.2. Determining the Optimal Number of Clusters (K)

Both the **Elbow Method** (Inertia) and the **Silhouette Score** were calculated for K values ranging from 2 to 10.

- *3.2.1 Elbow Method:*

Inertia measures the sum of the squared distances between each data point and the centroid (the average) of its assigned cluster. A lower inertia value means the clusters are more compact (denser and more cohesive). "We keep adding clusters to divide the data into more significant groups until adding more clusters doesn't make any difference."

- *3.2.2 Silhouette Score*

Evaluates both the **internal cohesion** and the **external separation** of the clusters. For every data point, the Silhouette Score calculates how similar the point is to its own cluster (cohesion) compared to the nearest neighboring cluster (separation).

The average score for all points ranges between **-1 and +1**:

- **High Score (close to +1):** Indicates the point is well-matched to its own cluster and well-separated from neighboring clusters.
- **Low Score (close to 0):** Indicates the point is very close to the boundary of another cluster, and the assignment is ambiguous.
- **Negative Score (close to -1):** Indicates the point has been assigned to the wrong cluster

Silhouette Scores:

    - K=2: **0.0983** (Highest score)
    - K=3: 0.0665
    - K=4: 0.0600
    - ...
- **Conclusion:** Based on the Silhouette Score, **K=2** is the mathematically best fit. However, analyzing the inertia plot and the business context, **K=3** was selected as it provides three distinct and meaningful groups for business strategy, despite the lower Silhouette Score.

3.3. Segmentation Results and Profiling

Once the K-Means algorithm has run and assigned a cluster label (0, 1, or 2) to each customer, the analyst's work begins. The goal here is to answer the question: Who are the customers in Cluster 0, 1, and 2, really?

Profiling takes these labels and cross-references them with the original customer features to define an archetype and give it a business name.

- We used the mean to analyze the numeric data. The mean is the best way to understand the **central or typical value** of a cluster.
- We used the mode for categorical data. The mode identifies the **most common or frequent category** within a cluster.

Profiling allows you to transform the abstract K-Means labels into business descriptions like:

- **Cluster 0:** "VIP Customers with High Average Spend"
- **Cluster 1:** "Rare Buyers with Low Spend"
- **Cluster 2:** "Active, Discount-Oriented Subscribers"

This step is critical because it enables marketing and product teams to develop **specific, profitable strategies** for each archetype, which is the ultimate goal of segmentation.

4. Predictive Modeling (Classification)

The goal of this stage was to predict a customer's **Purchase Segment** (Low, Medium, High) based on their features. We train supervised models that can learn the relationship between a customer's features and their **Purchase Segment** (the target variable).

4.1. Target Variable Creation

The original Purchase Amount (USD) is a **continuous variable**. This means it can take any value within a range (e.g., 45.00, 45.01, 72.99, etc.).

- If the goal were to predict the exact dollar value (e.g., $58.32), we would use a **Regression model**.
- Since the goal is to predict a category (Low, Medium, High), we must use a **Classification model**.

For the Classification model to work, we need to convert the continuous purchase amount into discrete categories.

- *4.1.2 Binning*

Binning is the process of taking a continuous variable and dividing its range of values into a limited number of intervals or "bins," transforming it into a **categorical variable**.

To divide the purchase amount, a method based on **percentiles** was used. Percentiles ensure that each new category has approximately the same number of customers.

- **33rd Percentile:** The purchase amount value below which 33% of all customers fall is calculated.
    - This value was **$45.00 USD**.

- **66th Percentile:** The purchase amount value below which 66% of all customers fall is calculated.
    - This value was **$73.00 USD**.

Using these two cutoff points, the continuous Purchase Amount (USD) variable was transformed into the new target variable Purchase Segment (Y):

Segment (Y)	Condition	Approximate Percentage of Customers
Low	Amount $\leq$ $45.00 USD	Approximately 33%
Medium	Amount between $45.01 USD and $73.00 USD	Approximately 33%
High	Amount $>$ $73.00 USD	Approximately 34%

The Purchase Segment variable is the Target Variable Y for the classification model. The model will learn to predict whether a new customer, based on their age, purchase history, location, etc., will fall into the Low, Medium, or High spending category.

4.2. Model Performance

Two classification models were trained and evaluated on the test set:

Model	**Overall Accuracy: % the model works**
Logistic Regression	51.41%
Random Forest Classifier	**49.36%**

Despite both models demonstrating **moderate performance** (around 50% accuracy), which indicates room for improvement, they nonetheless provide a **basic level of classification**. These results suggest that the relationship between the input features (including the segmentation cluster) and the three distinct Purchase Segments (Low, Medium, High) is complex and not fully explained by standard models.

- **Key Observation from the Classification Report:** For both models, the prediction performance for the **Medium** segment is significantly lower (low *precision* and *recall* values) compared to the Low and High segments. This fact suggests that the boundaries between Low/Medium and Medium/High purchase amounts appear to be the most challenging to predict based on the available features. **This is a key area for further investigation and model refinement.**

- *4.2.1 Why do we choose these models?*

1- Logistic Regression (Linear): **Simplicity and Speed.** It is the baseline model for any classification task. It is quick to train and gives an idea of the performance achievable with a simple, easy-to-interpret model (i.e., which features linearly influence the prediction).

2- Random Forest Classifier (Non-Linear): **Power and Robustness.** This is a benchmark model known for its robustness and its ability to capture complex, non-linear relationships between features. If this complex model cannot achieve high accuracy, it strongly suggests that the difficulty lies in the intrinsic nature of the data or the chosen features.

The standard practice is to **test one simple model and one complex model**. Since the complex model (Random Forest) did not significantly outperform the simple one (Logistic Regression), the conclusion is that the problem is inherently difficult to classify accurately with the current set of features, particularly around the "Medium" spending segment.

5. Conclusion and Application Potential

5.1. Key Conclusions

- **Effective Segmentation:** The K-Means clustering successfully identified three distinct, commercially relevant customer segments.
  - **VIP Customers (Cluster 0):** High spend and highest review ratings.
  - **Active Subscribers (Cluster 2):** Most numerous, subscribe to a service, and are responsive to discounts/promos.
  - **Rare Buyers (Cluster 1):** Lowest average spend and lowest purchase frequency (Annually).
- **Prediction Challenge:** The current feature set and models already provide a solid foundation for classification. The fact that predicting the exact segment (Low/Medium/High) requires additional focus highlights a significant potential for model improvement through further analysis of existing data or the inclusion of new variables.

5.2. Potential Applications of the Results

The core value lies in the **Customer Segmentation** and the ability to predict the **Cluster Name** for a new customer.

1. **Targeted Marketing:**
   - **Cluster 0 (VIP):** Focus on retention, loyalty programs, and offering exclusive, high-value products (as they value quality and spend highly).
   - **Cluster 2 (Active Subscribers):** Target with personalized discount offers and promotions to maximize immediate purchases and encourage cross-selling within the subscription.
   - **Cluster 1 (Rare Buyers):** Implement campaigns aimed at increasing their purchase frequency (e.g., small, frequent incentives) to move them toward a more active segment.

2. **Onboarding & Personalization:** When a new customer provides initial data (as demonstrated in the final code block), they can be immediately assigned to a **Cluster (e.g., VIP Customers)**. This allows the platform to tailor their initial experience (e.g., display high-value items for a VIP, show discounts for an Active Subscriber) to maximize the chance of the first purchase.